```
API Endpoints:
  - http://localhost:5000/

API Endpoints:


API Endpoints:


API Endpoints:
API Endpoints:
  - http://localhost:5000/
  - http://localhost:5000/api/docs
  - http://localhost:5000/api/health
  - http://localhost:5000/api/register (POST)
  - http://localhost:5000/api/verify (POST)
====================================================================
  - http://localhost:5000/
  - http://localhost:5000/api/docs
  - http://localhost:5000/api/health
  - http://localhost:5000/api/register (POST)
  - http://localhost:5000/
  - http://localhost:5000/api/docs
  - http://localhost:5000/
  - http://localhost:5000/api/docs
```

Cloud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

```
 * Debugger is active!
 * Debugger PIN: 744-248-591
 * Detected change in 'C:\\Users\\HP\\blockchain-iot-security\\src\\api_server_complete.py', reloadin
g
Exception in thread Thread-1 (serve_forever):
 * Restarting with stat
Initializing IoT Security System...
======================================================================
IOT BLOCKCHAIN SECURITY SYSTEM
======================================================================
Initializing system...

======================================================================
BLOCKCHAIN CLIENT INITIALIZATION
======================================================================
✓ Connected to blockchain: http://127.0.0.1:8545
✓ Account: 0xB45e8457a205749395496386 8CDA334Bf26B5014
✓ Balance: 0 ETH
✓ Smart contract loaded: 0xB15ECFFf17ed39A747CbCa45dE0442eaB3aE6B0B
======================================================================

✅ System initialized successfully!
======================================================================
 API Server Ready!

======================================================================
```

ud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

```
$ python src/api_server_complete.py

 * Debugger is active!
 * Debugger PIN: 744-248-591
 * Detected change in 'C:\\Users\\HP\\blockchain-iot-security\\src\\api_server_complete.py', reloadin
g
Exception in thread Thread-1 (serve_forever):
 * Restarting with stat
Initializing IoT Security System...
======================================================================
IOT BLOCKCHAIN SECURITY SYSTEM
======================================================================
Initializing system...

======================================================================
BLOCKCHAIN CLIENT INITIALIZATION
======================================================================
✓ Connected to blockchain: http://127.0.0.1:8545
✓ Account: 0xB45e8457a2057493954963868CDA334Bf26B5014
✓ Balance: 0 ETH
✓ Smart contract loaded: 0xB15ECFFf17ed39A747CbCa45dE0442eaB3aE6B0B
======================================================================

✅ System initialized successfully!
======================================================================
 API Server Ready!

======================================================================
IOT BLOCKCHAIN SECURITY API SERVER
```

oud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

Environmental Sensor Telemetry Data

155 | Code | Download

Data Card | Code (22) | Discussion (3) | Suggestions (0)

iot_telemetry_data.csv (61.93 MB)

Detail | Compact | Column

9 of 9 columns

About this file

Suggest Edits

Environmental telemetry data from a series of (3) IoT devices.

# ts	≙ device	# co	# humidity	✓ light
timestamp of reading	unique device name	carbon monoxide	humidity (%)	light detected?
1.59b – 1.6b	b8:27:eb:bf:9d:51 46% 00:0f:00:70:91:0a 28% Other (105919) 26%	0 – 0.01	1.1 – 99.9	true / false

Data Explorer

61.93 MB

iot_telemetry_data.csv

Summary

1 file

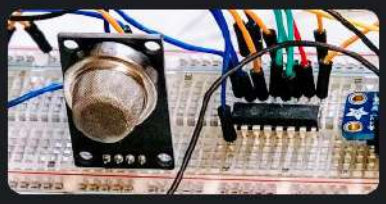
9 columns

GARY A. STAFFORD · UPDATED 6 YEARS AGO

155 Code Download

Environmental Sensor Telemetry Data

Temperature, humidity, CO, liquid petroleum gas (LPG), smoke, light, and motion



Data Card Code (22) Discussion (3) Suggestions (0)

About Dataset

Context

Environmental sensor telemetry data, detailed in the blog post, Getting Started with IoT Analytics on AWS, published on Towards Data Science.

Content

The data was generated from a series of three identical, custom-built, breadboard-based sensor arrays. Each array was

Usability
10.00

License
CC0: Public Domain

Expected update frequency
Never

Search

Environmental Sensor Telemetry Data

155 | Code | Download

Data Card | Code (22) | Discussion (3) | Suggestions (0)

Detail | Compact | Column

9 of 9 columns

# ts timestamp of reading	device unique device name	# co carbon monoxide	# humidity humidity (%)	light light detected?
1.59b 1.6b	b8:27:eb:bf:9d:51 46% 00:0f:00:70:91:0a 28% Other (105918) 26%	0 0.01	1.1 99.9	
1.5945120943859746E9	b8:27:eb:bf:9d:51	0.004955938648391245	51.0	false
1.5945120947355676E9	00:0f:00:70:91:0a	0.0028410088607101570 6	76.0	false
1.5945120980735729E9	b8:27:eb:bf:9d:51	0.004976012340421658	50.9	false
1.5945120999589146E9	1c:bf:ce:15:ec:4d	0.004408026829699689	76.80000305175781	true
1.5945121017612355E9	b8:27:eb:bf:9d:51	0.004967363641908952	50.9	false

iot_telemetry_data.csv

Summary

1 file

9 columns

Content

The data was generated from a series of three identical, custom-built, breadboard-based sensor arrays. Each array was connected to a Raspberry Pi devices. Each of the three IoT devices was placed in a physical location with varied environmental conditions.

device	environmental conditions
00:0f:00:70:91:0a	stable conditions, cooler and more humid
1c:bf:ce:15:ec:4d	highly variable temperature and humidity
b8:27:eb:bf:9d:51	stable conditions, warmer and dryer

Each IoT device collected a total of seven different readings from the four sensors on a regular interval. Sensor readings include temperature, humidity, carbon monoxide (CO), liquid petroleum gas (LPG), smoke, light, and motion. The data spans the period from **07/12/2020 00:00:00 UTC – 07/19/2020 23:59:59 UTC**. There is a total of **405,184** rows of data.

The sensor readings, along with a unique device ID and timestamp, were published as a single message, using the ISO standard Message Queuing Telemetry Transport (MQTT) network protocol. Below is an example of an MQTT message payload.

Expected update frequency

Never

Tags

Earth and Nature

Electronics

Oil and Gas

```
(blockchain_env)
HP@DESKTOP-FCU03R7 MINGW32 ~/blockchain-iot-security
$ python src/api_test_script.py

======================================================================
IoT BLOCKCHAIN SECURITY API - TEST SUITE
======================================================================

[TEST] Home endpoint
  Method: GET
  URL: http://localhost:5000/

  Status Code: 200
  Response: {
  "author": "Oyelade Paul Oluwafemi",
  "documentation": "/api/docs",
  "endpoints": {
    "/api/audit": "GET - Get system audit report",
    "/api/hash": "POST - Generate hash for data",
    "/api/health": "GET - System health check",
    "/api/records": "GET - Get total blockchain records",
    "/api/register": "POST - Register IoT data on blockchain",
    "/api/stats": "GET - Get system statistics",
    "/api/verify": "POST - Verify data integrity"
  },
  "service": "IoT Blockchain Security System API",
  "version": "1.0.0"
}

[TEST] API documentation
  Method: GET
```

Cloud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

```
$ source blockchain_env/Scripts/activate
(blockchain_env)
HP@DESKTOP-FCU03R7 MINGW32 ~/blockchain-iot-security
$ ganache-cli --port 8545 --networkId 1337 --accounts 10
This version of µWS is not compatible with your Node.js build:

Error: Cannot find module '../binaries/uws_win32_x64_127.node'
Require stack:
- C:\Users\HP\AppData\Roaming\npm\node_modules\ganache\node_modules\@trufflesuite\uws-js-unofficial\s
rc\uws.js
- C:\Users\HP\AppData\Roaming\npm\node_modules\ganache\dist\node\cli.js
Falling back to a NodeJS implementation; performance may be degraded.


ganache v7.9.2 (@ganache/cli: 0.10.2, @ganache/core: 0.10.2)
Starting RPC server

Available Accounts
==================
(0) 0x2E181C37a9B601802Ae7B90B3A10c99C8D172142 (1000 ETH)
(1) 0xeEE1bD6BEE8e32645F83D6238dee7BfC33615F80 (1000 ETH)
(2) 0x5b198bbD70FEF702062003E4d7AFB02d819792b0 (1000 ETH)
(3) 0xb063d08B9e157c8c1836A8152446be068DA189C4 (1000 ETH)
(4) 0xB3FfFBC69600568B9463855D67A7e6f27dD9aa5D (1000 ETH)
(5) 0x5551404d0cF8342486F3022cd6E5eD6bDa18251F (1000 ETH)
(6) 0x54d8dF8666f69c4b768A9f4B350322B26280908F (1000 ETH)
(7) 0x8fA10419EAA07dE49Ad887c2D64a6E7712eF601F (1000 ETH)
(8) 0x94e1170211040749CE56A9C1B38776Dd001FF376 (1000 ETH)
```

Cloud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

```
(blockchain_env)
HP@DESKTOP-FCU03R7 MINGW32 ~/blockchain-iot-security
$ ganache-cli --port 8545 --networkId 1337 --accounts 10
Hardfork: shanghai
Id:       1337


Chain
Chain
==================
Hardfork: shanghai
Id:       1337

RPC Listening on 127.0.0.1:8545
web3_clientVersion
eth_getBalance
web3_clientVersion
eth_getBalance
> web3_clientVersion
eth_getBalance
eth_chainId
eth_call
eth_chainId
eth_getCode
web3_clientVersion
eth_getBalance
eth_chainId
eth_call
eth_chainId
eth_getCode
eth_chainId
```

oud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live

```
HP@DESKTOP-FCU03R7 MINGW32 ~/blockchain-iot-security
$ python src/api_server_complete.py
Initializing IoT Security System...
======================================================================
IOT BLOCKCHAIN SECURITY SYSTEM
======================================================================
Initializing system...

======================================================================
BLOCKCHAIN CLIENT INITIALIZATION
======================================================================
✓ Connected to blockchain: http://127.0.0.1:8545
✓ Account: 0xB45e8457a2057493954963868CDA334Bf26B5014
✓ Balance: 0 ETH
✓ Smart contract loaded: 0xB15ECFFf17ed39A747CbCa45dE0442eaB3aE6B0B
======================================================================

✅ System initialized successfully!
======================================================================
✅ API Server Ready!

======================================================================
IOT BLOCKCHAIN SECURITY API SERVER
======================================================================
Server starting on 0.0.0.0:5000
Blockchain: http://127.0.0.1:8545
Contract: 0xB15ECFFf17ed39A747CbCa45dE0442eaB3aE6B0B
```

Cloud Code - No Project Ln 371, Col 17 Spaces: 4 UTF-8 LF { } Python 3.13.2 (blockchain_env) Go Live